



PRAVEEN KUMAR OJHA

Process & Data Analyst

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Scan for Portfolio

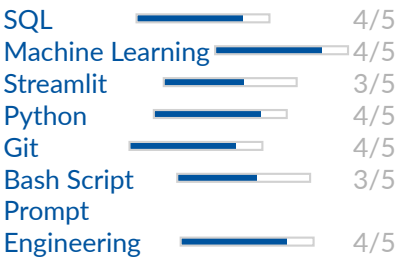
MOTIVATION

Data Analyst with 4+ years of experience in Process Mining and Data Engineering at Bosch. Skilled in Celonis optimization, LLM-based conformance checking, and hands-on machine learning projects. Currently blending Analytics with process intelligence as part of my master thesis. Seeking roles where I can apply **data-driven thinking** to **optimize** and **automate** business processes.

Let's connect if you're solving complex process challenges or need sharp, scalable data analysis expertise!

TECHNICAL SKILLS

CORE EXPERTISE



TECHNOLOGY LANDSCAPE

- Optimization:** Heuristic Opt., gurobipy, cplex
- ML & Viz:** scikit-learn, tensorflow, seaborn
- Infra & Web:** Flask, Docker, Grafana, Shell, pyodbc

PROCESS KNOWLEDGE

- O2C (Order-to-Cash)
- B2C (Bill-to-Cash)
- P2P (Procure-to-Pay)
- A2A (Affiliate-to-Buisness)
- I2P (Invoice-to-Pay)

WORK EXPERIENCE

Master Thesis

03/2025 - 07/2025

Bosch

Stuttgart, Germany

- Automated benchmarking of 10+ Large Language Models (LLMs) for BPMN activity mapping and conformance checking using a modular Python tool, streamlining integration across 7 REST APIs.
- Benchmarked LLM mapping accuracy against manually validated gold standards, delivering advanced evaluation metrics (precision, recall, F1) and achieving 20% improvement in workflow processing speed.
- Enabled end-to-end process mining conformance analysis by linking LLM-generated mappings with token-based replay metrics, supporting dynamic dashboards and business-ready validation.

Tech Stack: Python Streamlit REST API LLM pm4py pycelonis

Data Engineering in Process Mining Intern

09/2024 - 02/2025

Bosch

Stuttgart, Germany

- Streamlined platform tasks (Data Pool backups, job health checks) with 60% faster execution, and automated regex-based outlier detection for access anomalies, cutting manual audit effort by 80%.
- Designed Streamlit application which automated process conformance checks; enabled structured activity mapping workflows used in 3+ pilot use cases during the PoC phase.
- Launched a Gamified leader board using LLM-scored commits to boost versioning, increasing developer engagement by 45% and automated weekly adoption tracking via Celonis ML Workbench and Action Flow.

Tech Stack: Python PyCelonis SQL Git / Gitflow Streamlit

Working Student

11/2023 - 08/2024

Bosch

Stuttgart, Germany

- Optimized Celonis transformation queries and data model loading, cutting use case runtime and memory usage by 40%.
- Built automated permission reporting dashboard, enabling quick detection of false permissions and improving reporting accuracy by 75%.

Tech Stack: Python Celonis SQL Action Flows

Senior Associate Consultant

09/2019 - 08/2023

Bosch Global Software Solutions

Bangalore, India

- Led B2C process performance tuning, reducing runtime from 24 to 10 hours for 32M data through strategic query and memory optimization.
- Orchestrated process flow transition from Sales Order to Purchase Order, covering 3.5M records across 7 systems from a 13-month dataset.
- Supported AR use case setup, uploading a 30M-record data model from 7 systems via Oracle SQL to SAP HANA, enabling large-scale process analysis.

Tech Stack: SQL Celonis B2C P2P I2P A2A IBC ETL

SOFT SKILLS

Agile Methodology

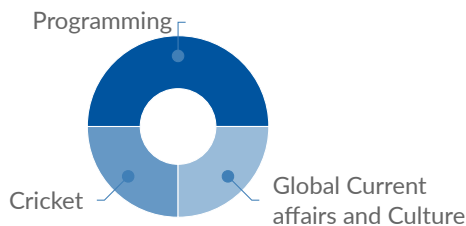
Process Consultation

Problem-Solving

LANGUAGES

Hindi Native
English C1 (Professional)
German A1 (Basic)

INTEREST AREAS



Data Analyst Intern

📅 2018

Tata Steel

📍 Jamshedpur, India

- **Developing Employee Turnover Prediction Model** - Developed a predictive model for employee turnover, identifying key attrition factors and helping HR design targeted retention strategies.

Tech Stack: Python Pandas Scikit-learn Matplotlib

EDUCATION

M.Sc. in Data Analytics and Decision Science

RWTH Aachen University

🎓 GPA - 1,5

📅 10/2023 – 09/2025

CO₂ EMISSIONS FORECASTING FOR EU COUNTRIES

ML ensemble prediction of EU carbon emissions

🔗 Repository

- Developed a machine learning pipeline to **predict national CO₂ emissions** across EU countries by extending the methodology from a [2024 Scientific Reports paper](#).
- Benchmarked ML models (GBM, SVR, LSTM, Ensemble) across 10-year horizons in a Dockerized pipeline with Grafana dashboard and PostgreSQL-backed experimentation.

Tech Stack: Tensorflow Scikit-learn Seaborn Grafana Docker

OPTIMIZATION OF TRANSPORT PLANNING AND INBOUND PROCESSING IN PACKAGE CENTER

Routing and scheduling for overnight delivery

🔗 Repository

- This project focuses on optimizing the parcel transportation network of DHL Deutsche Post, which involves overnight delivery across 37 centers.
- The optimization problem involves determining the departure times, routes, and loading strategies for swap bodies, while adhering to constraints such as sorting capacities, cut-off times, and driving regulations

Tech Stack: Python Gurobi Cplex Streamlit Git

ECOFORCAST: REVOLUTIONIZING GREEN ENERGY SURPLUS PREDICTION IN EUROPE

Hackathon

- We were asked to develop a model to predict the country with the highest surplus of green energy in the upcoming hour among nine specified countries. Thus, considering renewable energy generation (e.g., wind, solar, geothermal) and energy consumption to calculate the surplus.
- Ranked 13th out of 237 teams in the hackathon challenge.

Tech Stack: Python Pandas NumPy scikit-learn Matplotlib

B.Tech. in Computer Science and Engineering

SOA University, India

🎓 CGPA: 9.83/10 (GPA: 1,1)

📅 07/2015 – 06/2019

VEHICLE SPEED DETECTION FROM THE CAPTURED VIDEO

Computer Vision

- Developed computer-vision software to identify speeding vehicles from highway camera footage, providing automated violation detection and improving highway safety monitoring.

Tech Stack: Python OpenCV